KINGDOM OF CORPUS · STUDY NOTES

Land 3: Tessera Marsh

The four tissue types · OpenStax Ch. 4

Key ideas

Four Tissues and Germ Layers

- The four primary tissues are epithelial, connective, muscle and nervous.
- Ectoderm forms the epidermis and the nervous system.
- Mesoderm forms muscle, bone, blood and most connective tissue.
- Endoderm forms the epithelial lining of the digestive and respiratory tracts and many glands.
- Histology is the study of tissues.

Epithelial Traits

- Epithelium covers surfaces and lines cavities and is packed tightly with little matrix.
- It is avascular and is fed by diffusion from the connective tissue beneath it.
- It shows polarity: an apical surface facing a space and a basal surface on the basement membrane.
- The basement membrane (basal lamina plus reticular lamina) anchors epithelium to connective tissue.
- Stem cells give epithelium a high capacity to regenerate.

Naming Epithelium

- Name by layers first (simple or stratified), then by the shape of the surface cells (squamous, cuboidal, columnar).
- Simple squamous lines alveoli, blood vessels (endothelium) and serous membranes (mesothelium) for diffusion and filtration.
- Simple cuboidal lines kidney tubules and gland ducts; simple columnar lines the stomach and intestines.
- Pseudostratified ciliated columnar with goblet cells lines the trachea; all cells touch the basement membrane.
- Stratified squamous protects: keratinized in the epidermis, nonkeratinized in the mouth, esophagus and vagina.
- Transitional epithelium stretches in the urinary bladder and ureters.

Cell Junctions

- Tight junctions seal neighboring cells so fluid cannot leak between them, as in the gut lining.
- Desmosomes rivet cells together against mechanical stress in the epidermis and heart.
- Hemidesmosomes anchor cells to the basement membrane.
- Gap junctions are connexon channels that pass ions and allow cardiac and smooth muscle to act in unison.

Glands

- Endocrine glands are ductless and release hormones into the blood.
- Exocrine glands release products through ducts onto a surface; goblet cells are unicellular exocrine glands.
- Merocrine secretion uses exocytosis and the cell stays intact (salivary, eccrine sweat glands).
- Apocrine secretion pinches off the apical part of the cell with the product.
- Holocrine secretion ruptures the whole cell, as in sebaceous glands.

Connective Tissue

- Connective tissue is cells plus matrix; the matrix is protein fibers plus ground substance.
- Fibroblasts make the matrix; adipocytes store fat; macrophages engulf debris; mast cells release histamine and heparin.
- Collagen fibers give strength, elastic fibers allow stretch and recoil, reticular fibers form a supportive mesh.
- Loose tissues: areolar (under epithelia), adipose (fat storage, insulation), reticular (stroma of spleen and lymph nodes).
- Dense regular tissue forms tendons and ligaments; dense irregular forms the dermis.
- Supportive tissues: hyaline cartilage (joint surfaces, growth plates), elastic cartilage (ear, epiglottis), fibrocartilage (discs, menisci), bone; blood is fluid connective tissue.

Muscle and Nervous Tissue

- Skeletal muscle: long, cylindrical, striated, multinucleated, voluntary.
- Cardiac muscle: striated, branched, one or two central nuclei, intercalated discs, involuntary.
- Smooth muscle: spindle shaped, one nucleus, no striations, involuntary, in walls of hollow organs and vessels.
- Neurons conduct impulses: dendrites receive signals and the axon carries impulses away from the cell body.
- Neuroglia support and protect neurons and can still divide.

Membranes

- Mucous membranes line cavities open to the outside, such as the digestive, respiratory and urinary tracts.
- Serous membranes line closed cavities (pleura, pericardium, peritoneum) and secrete slippery serous fluid.
- The cutaneous membrane is the skin, a dry membrane.
- Synovial membranes line joint cavities and are connective tissue only, with no epithelium.

Inflammation, Repair, Aging and Cancer

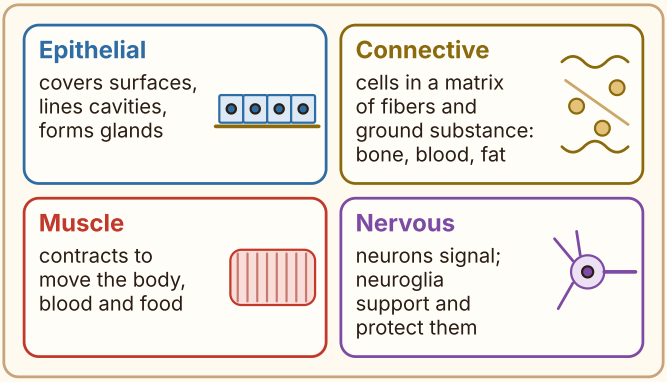
- Cardinal signs of inflammation: redness, heat, swelling and pain, often with loss of function.
- Histamine from mast cells dilates vessels and raises capillary permeability, causing redness, heat and swelling.
- Regeneration replaces tissue with the same cell type; fibrosis replaces it with collagen scar made by fibroblasts.
- Epithelium and bone regenerate well; cardiac muscle and most nervous tissue heal by scarring.
- With age, elastic fibers decline, collagen stiffens and repair slows.
- Benign tumors stay local; malignant tumors invade and can metastasize through blood or lymph.

Villager wisdom

- Old Tomas the weaver says to name an epithelium by its layers first, then the shape of the top cells: simple or stratified, then squamous, cuboidal or columnar.
- Epithelium has no blood vessels of its own. It drinks by diffusion from the connective tissue beneath it, like moss growing on a damp log.
- Every connective tissue is cells living in a matrix of fibers and ground substance. Even blood counts, because its matrix is liquid plasma.
- To tell muscle apart, look for stripes and branches: skeletal is striped and long, cardiac is striped and branched with intercalated discs, and smooth has no stripes at all.
- Redness, heat, swelling and pain mean inflammation is at work. Histamine opens the vessels and makes them leaky so help can reach the wound.

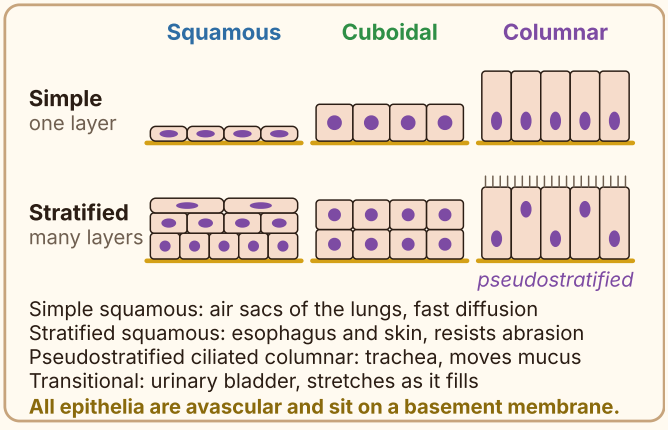
Diagrams

The four tissue types



All body tissues fall into four types: epithelial covers and lines, connective supports with cells in a matrix, muscle contracts, and nervous tissue signals with neuroglia in support.

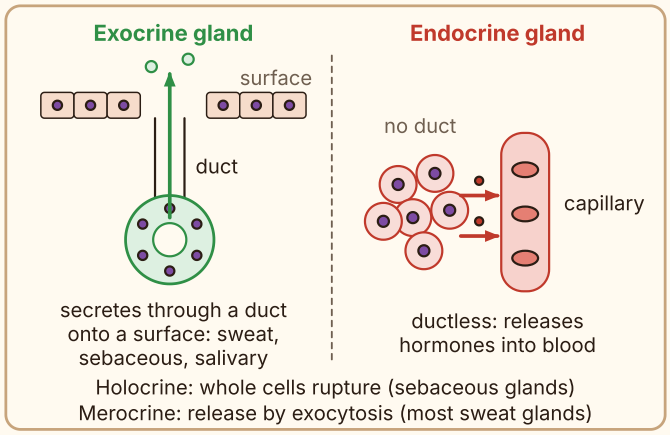
Classifying epithelium



Simple squamous: air sacs of the lungs, fast diffusion
Stratified squamous: esophagus and skin, resists abrasion
Pseudostratified ciliated columnar: trachea, moves mucus
Transitional: urinary bladder, stretches as it fills
All epithelia are avascular and sit on a basement membrane.

Epithelium is named by layers (simple or stratified) and by the shape of the surface cells (squamous, cuboidal, columnar); form matches function.

Exocrine and endocrine glands



Exocrine glands release secretions through ducts onto a surface; endocrine glands are ductless and release hormones into the blood.

Connective tissue types

Cells scattered in an extracellular matrix

Connective tissue proper

Loose	Dense
Areolar: under epithelia	Regular: tendons, ligaments
Adipose: stores fat	Irregular: dermis
Reticular: mesh in spleen	Elastic: large artery walls

Supportive connective tissue

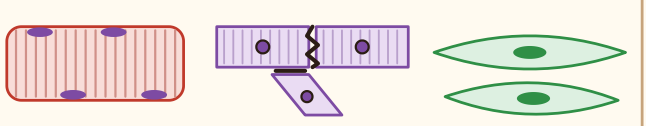
Cartilage (avascular)	Bone
Hyaline: growth plates	Osteocytes in lacunae
Fibrocartilage: discs	Calcified, rigid matrix
Elastic: external ear	

Fluid: blood, lymph

Fibroblasts make the fibers: collagen (strongest), elastic, reticular

All connective tissues are cells scattered in an extracellular matrix of ground substance and fibers; cartilage is avascular and tendons have few vessels, so both heal slowly.

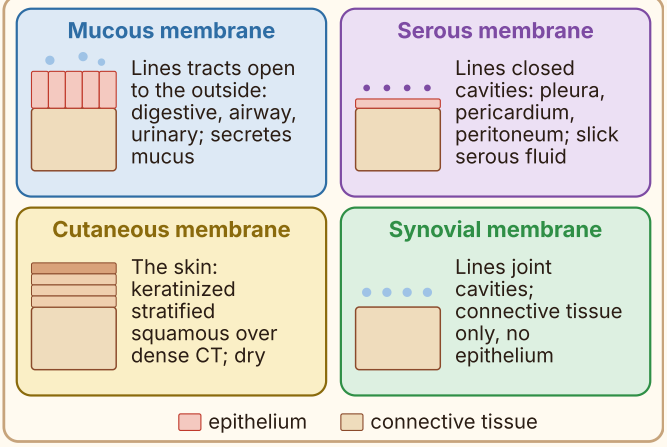
Three types of muscle tissue



Skeletal	Cardiac	Smooth
striated	striated, branched	no striations
many nuclei	intercalated discs	one central nucleus
voluntary	involuntary	involuntary
on bones	**heart wall**	**vessel and organ walls**

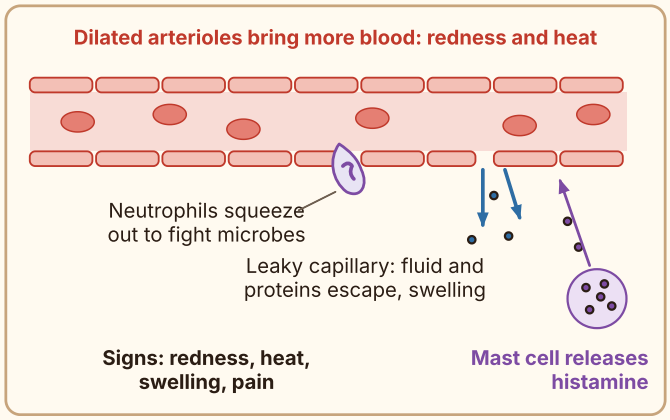
Skeletal muscle is striated, multinucleate and voluntary; cardiac muscle is striated, branched and joined by intercalated discs; smooth muscle is spindle shaped with no striations.

Tissue membranes



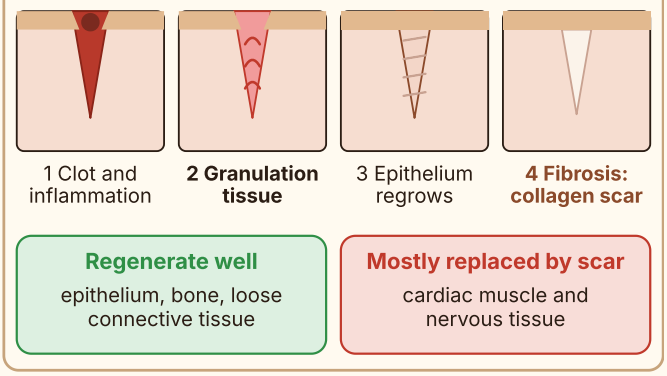
Mucous, serous and cutaneous membranes are epithelium over connective tissue; the synovial membrane lining joint cavities is connective tissue only, with no epithelium.

Inflammation



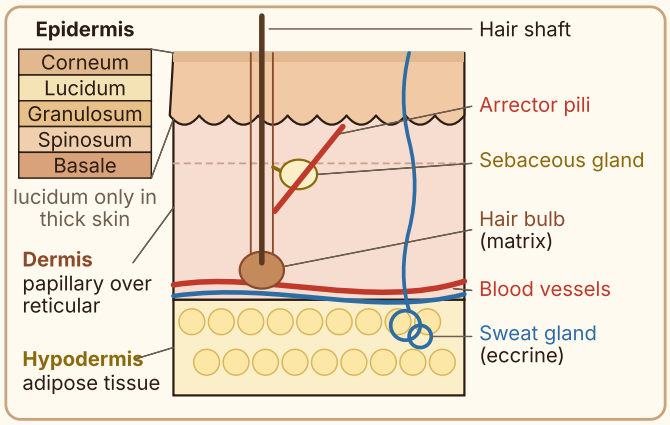
Histamine dilates arterioles (redness, heat) and makes capillaries leaky, so fluid and proteins escape (swelling) and neutrophils move into the tissue.

Tissue repair



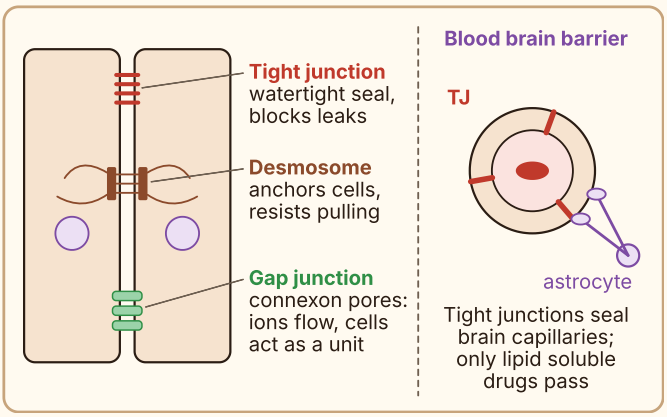
A deep wound fills with a clot, then vascular granulation tissue; fibroblasts lay down collagen that becomes a scar (fibrosis), the usual outcome where tissue such as cardiac muscle cannot regenerate.

Layers of the skin



The epidermis is stratified squamous epithelium renewed from the stratum basale; the dermis holds glands, follicles and vessels; the fatty hypodermis insulates and cushions.

Cell junctions and the blood brain barrier



Tight junctions seal neighboring cells, desmosomes rivet them against pulling, and gap junctions (connexons) let ions pass so cells act together; tight junctions in brain capillaries form the blood brain barrier.

Self check

1. **Loose areolar connective tissue is found:**
    A. Beneath epithelia, binding them to deeper tissue
    B. In tendons attaching muscle to bone
    C. In the external ear
    D. As the framework of the skeleton
2. **Redness and heat at an inflamed site are caused mainly by:**
    A. Vasoconstriction lowering blood flow
    B. Fibroblasts forming scar tissue
    C. Loss of fluid from the tissue
    D. Vasodilation raising local blood flow
3. **A biopsy shows branched, striated cells joined by intercalated discs. The sample came from the:**
    A. Stomach wall
    B. Heart
    C. Biceps brachii
    D. Urinary bladder
4. **Which cell is the most common in connective tissue proper and produces its fibers?**
    A. Adipocyte
    B. Macrophage
    C. Mast cell
    D. Fibroblast
5. **Which epithelium allows the urinary bladder to stretch as it fills?**
    A. Transitional
    B. Simple squamous
    C. Pseudostratified columnar
    D. Stratified cuboidal
6. **Which cell junction forms a watertight seal between neighboring epithelial cells?**
    A. Tight junction
    B. Desmosome
    C. Gap junction
    D. Hemidesmosome
7. **Epithelial tissue has no blood vessels of its own. How do its cells get nutrients?**
    A. From capillaries running between the cells
    B. By diffusion from underlying connective tissue
    C. From nerve endings within the tissue
    D. By storing glucose inside keratin

8. **Goblet cells, which secrete mucus onto an epithelial surface, are an example of a:**
    A. Unicellular endocrine gland
    B. Unicellular exocrine gland
    C. Multicellular endocrine gland
    D. Holocrine sebaceous gland
9. **Why is simple squamous epithelium ideal for lining the air sacs of the lungs?**
    A. It secretes large amounts of mucus
    B. It resists abrasion with many layers
    C. Its thin single layer allows fast diffusion
    D. It stretches greatly as air enters
10. **Which cells in nervous tissue support and protect neurons but do not conduct impulses?**
    A. Fibroblasts
    B. Myocytes
    C. Neuroglia
    D. Chondrocytes
11. **A radiograph of a 10 year old shows growth plates in the long bones. These plates are made of:**
    A. Fibrocartilage
    B. Elastic cartilage
    C. Hyaline cartilage
    D. Dense regular tissue
12. **A patient has a herniated disc. The tough ring of the intervertebral disc, which absorbs shock between vertebrae, is made of:**
    A. Hyaline cartilage
    B. Fibrocartilage
    C. Elastic cartilage
    D. Areolar tissue

Answer key

1. **A.** Beneath epithelia, binding them to deeper tissue. Areolar tissue fills spaces under epithelia and around organs, cushioning them and holding tissue fluid. Tendons are dense regular connective tissue, which is packed with parallel collagen.
2. **D.** Vasodilation raising local blood flow. Mediators such as histamine dilate arterioles, so more warm blood flows into the area. Vasoconstriction would reduce blood flow and make the area pale and cool.
3. **B.** Heart. Branching cells with intercalated discs are unique to cardiac muscle, which lets the heart contract as a coordinated unit. Skeletal muscle like the biceps is striated too, but its cells are long, unbranched and lack intercalated discs.
4. **D.** Fibroblast. Fibroblasts secrete collagen, elastic and reticular fibers and the ground substance. Macrophages also live in connective tissue, but they engulf debris and microbes rather than build the matrix.
5. **A.** Transitional. Transitional epithelium has cells that change from rounded to flattened as the bladder wall stretches. Simple squamous is thin but is a single fragile layer that could not protect against urine.
6. **A.** Tight junction. Tight junctions fuse adjacent plasma membranes so fluid cannot leak between cells. Desmosomes hold cells together against pulling forces but do not seal the gaps.

7. **B.** By diffusion from underlying connective tissue. Epithelium is avascular, so nutrients diffuse up from capillaries in the connective tissue beneath the basement membrane. Epithelium is innervated, but nerves carry signals, not nutrients.
8. **B.** Unicellular exocrine gland. A goblet cell is a single cell that releases mucus onto a surface, making it a unicellular exocrine gland. It is not endocrine because its product is not released into the blood as a hormone.
9. **C.** Its thin single layer allows fast diffusion. Flat cells in a single layer give oxygen and carbon dioxide the shortest possible path for diffusion. Resistance to abrasion comes from stratified epithelium, which would slow gas exchange.
10. **C.** Neuroglia. Neuroglia (glial cells) insulate, nourish and protect neurons, while neurons themselves carry electrical signals. Fibroblasts make fibers in connective tissue and are not part of nervous tissue.
11. **C.** Hyaline cartilage. The epiphyseal growth plates are hyaline cartilage, the most abundant cartilage, which also covers joint surfaces. Fibrocartilage is tougher and found in intervertebral discs and the pubic symphysis, not growth plates.
12. **B.** Fibrocartilage. Fibrocartilage has thick collagen bundles that resist compression and tension, ideal for intervertebral discs and the knee menisci. Hyaline cartilage is smoother and weaker and lines joint surfaces instead.

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